Practical 2f: Exception Handling

AIM: To demonstrate exception handling in C# using try-catch-finally blocks, custom exceptions, and multiple catch handlers.

PROGRAM CODE

```
using System;

namespace ExceptionHandling
{
   // Custom exception
   class InvalidAgeException : Exception
   {
      public InvalidAgeException(string msg) : base(msg) { }
   }

   class Program
   {
      static void ValidateAge(int age)
      {
         if (age < 0 || age > 120)
            throw new InvalidAgeException($"Invalid age: {age}");
      }

      static void Main(string[] args)
      {
         // 1. DivideByZeroException
         Console.WriteLine("--- DivideByZero Exception ---");
         try
         {
            int a = 10, b = 0;
            Console.WriteLine(a / b);
         }
         catch (DivideByZeroException ex)
         {
            Console.WriteLine($"Caught: {ex.Message}");
         }
         finally
         {
```

```
            Console.WriteLine("Finally block executed.");
        }

        // 2. FormatException
        Console.WriteLine("\n--- Format Exception ---");
        try
        {
            int num = int.Parse("abc");
        }
        catch (FormatException ex)
        {
            Console.WriteLine($"Caught: {ex.Message}");
        }

        // 3. IndexOutOfRange
        Console.WriteLine("\n--- IndexOutOfRange Exception ---");
        try
        {
            int[] arr = { 1, 2, 3 };
            Console.WriteLine(arr[10]);
        }
        catch (IndexOutOfRangeException ex)
        {
            Console.WriteLine($"Caught: {ex.Message}");
        }

        // 4. Custom Exception
        Console.WriteLine("\n--- Custom Exception ---");
        try
        {
            ValidateAge(-5);
        }
        catch (InvalidAgeException ex)
        {
            Console.WriteLine($"Custom Exception: {ex.Message}");
        }

        Console.ReadKey();
    }
}
```

}

OUTPUT

```
--- DivideByZero Exception ---
Caught: Attempted to divide by zero.
Finally block executed.

--- Format Exception ---
Caught: Input string was not in a correct format.

--- IndexOutOfRange Exception ---
Caught: Index was outside the bounds of the array.

--- Custom Exception ---
Custom Exception: Invalid age: -5
```